

NAME:
NEPTUN:

First Solution:
Second Solution:

1. Suppose that discount factors for different maturities (one, two, and three years) are given as 92%, 85%, and 79%, respectively. Today a financial institution entered into a long interest rate swap with company X. Interest payments are exchanged once a year. Under the terms of the swap, they exchange a fixed rate for the one-year LIBOR on a principal of \$500 million for two years. What is the first net CF of the swap for a financial institution?

- a) $-42,37 \text{ million \$} + \text{Libor}_1(0) * 500 \text{ million \$}$
- b) $+42,37 \text{ million \$} - \text{Libor}_1(0) * 500 \text{ million \$}$
- c) $-41,01 \text{ million \$} + \text{Libor}_1(0) * 500 \text{ million \$}$
- d) $+41,01 \text{ million \$} - \text{Libor}_1(0) * 500 \text{ million \$}$

Long IRS - paying fix receiving floating rate

par yield = $(1 - 0,92) / (0,92 + 0,85) = 8,47\%$

net CF1 = $-8,47\% * 500 + \text{Libor}_1(0) * 500$

net CF1 = $-42,37 \text{ million \$} + \text{Libor}_1(0) * 500 \text{ million \$}$

2. A currency swap has a remaining life of 20 months. It involves exchanging interest at 3% on 10 million euros for interest at 4% on 11 million dollars once a year. How can the party paying euro replicate the CCS with forward contracts?

- a) $\text{SF}_{\text{EURUSD}}(0,67) + \text{SF}_{\text{EURUSD}}(1,67)$
- b) $\text{LF}_{\text{EURUSD}}(0,67) + \text{LF}_{\text{EURUSD}}(1,67)$
- c) $\text{SF}_{\text{EUR}}(0,67) + \text{LF}_{\text{USD}}(1,67)$
- d) $\text{SF}_{\text{USD}}(0,67) + \text{LF}_{\text{EUR}}(1,67)$

base currency EUR, we sell the underlying asset SF position

$\text{CCS}(1,67) = \text{SF}_{\text{EURUSD}}(0,67) + \text{SF}_{\text{EURUSD}}(1,67)$